

FAIRBANKS, AK, USA

WMO: 702610

Lat: 64.804N

Lon: 147.876W

Elev: 432

StdP: 14.47

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26411

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-42.2	-37.6	-34.7	0.8	-28.2	-32.7	0.9	-27.0	17.7	6.5	14.5	4.6	0.5	20	0.663

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.7	80.8	60.9	77.7	59.9	74.5	58.4	63.3	76.5	61.7	73.5	60.1	70.8	5.7	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.8	75.1	65.6	57.0	70.5	64.3	55.4	66.3	63.2	28.9	76.2	27.7	73.3	26.6	70.8	72.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.6	13.6	11.7	DB	-44.5	86.4	4.8	3.0	-47.9	88.6	-50.7	90.4	-53.4	92.0	-56.8	94.2	(4)
(5)			WB	-32.9	66.6	2.0	2.5	-34.3	68.4	-35.5	69.8	-36.6	71.2	-38.1	73.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	28.6	-8.1	1.0	10.8	33.9	50.4	60.9	62.8	57.0	46.3	26.6	4.1	-4.5	(6)
(7)		DBStd	28.22	17.97	14.99	13.26	10.67	7.87	5.69	4.95	5.90	7.19	11.28	13.92	14.99	(7)
(8)		HDD50	8917	1802	1371	1215	486	89	2	0	8	150	726	1378	1690	(8)
(9)		HDD65	13366	2267	1791	1680	934	454	148	100	258	561	1190	1828	2155	(9)
(10)		CDD50	1094	0	0	0	2	103	328	396	225	39	1	0	0	(10)
(11)		CDD65	67	0	0	0	0	2	24	31	10	0	0	0	0	(11)
(12)		CDH74	764	0	0	0	0	57	285	312	109	1	0	0	0	(12)
(13)		CDH80	118	0	0	0	0	6	51	44	17	0	0	0	0	(13)
(14)	Wind	WSAvg	3.9	2.0	2.7	4.3	5.5	5.9	5.3	4.8	4.5	4.3	3.6	2.4	1.8	(14)
(15)	Precipitation	PrecAvg	13.2	0.6	0.5	0.4	0.4	0.7	1.8	2.8	2.3	1.6	0.7	0.7	0.7	(15)
(16)		PrecMax	20.2	2.8	1.7	2.4	2.9	2.1	3.3	5.9	3.7	3.3	1.4	1.9	3.1	(16)
(17)		PrecMin	9.1	0.1	0.1	0.0	0.0	0.2	0.5	0.1	0.4	0.3	0.2	0.1	0.0	(17)
(18)		PrecStd	2.5	0.6	0.4	0.5	0.6	0.5	0.7	1.4	0.9	0.7	0.3	0.6	0.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.4	36.1	47.1	66.0	79.9	85.4	84.7	82.1	70.4	57.7	36.0	34.1	(19)
(20)			MCWB	31.9	31.6	36.9	48.8	55.8	61.6	63.4	63.5	54.6	46.7	32.6	31.1	(20)
(21)		2%	DB	27.7	31.7	40.9	59.3	74.4	80.7	80.4	77.1	65.9	49.4	32.2	27.4	(21)
(22)			MCWB	25.9	29.4	33.4	44.4	54.1	59.9	61.9	61.8	52.4	41.8	30.8	25.6	(22)
(23)		5%	DB	22.6	26.6	36.1	54.7	70.0	77.2	77.5	72.0	62.5	44.5	28.0	22.2	(23)
(24)			MCWB	21.5	24.6	30.7	42.1	52.2	58.5	60.8	58.9	51.1	39.4	26.6	21.1	(24)
(25)		10%	DB	17.0	21.7	31.3	50.6	65.9	73.8	74.6	68.1	58.8	40.2	23.4	17.2	(25)
(26)			MCWB	16.1	20.4	27.0	39.9	50.0	57.2	59.7	57.2	49.6	36.6	22.3	16.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.0	33.0	37.6	49.3	57.6	64.3	65.9	55.7	47.8	33.7	31.3	(27)	
(28)			MDCB	36.1	34.9	45.7	64.4	74.5	80.7	78.7	78.4	66.7	55.4	35.2	33.6	(28)
(29)		2%	WB	26.8	29.3	34.3	45.6	55.2	61.9	63.7	62.6	53.5	42.5	30.8	26.0	(29)
(30)			MDCB	28.6	31.9	40.1	58.1	71.6	76.3	76.7	73.5	63.1	48.0	32.2	27.7	(30)
(31)		5%	WB	22.4	24.9	30.8	42.4	53.2	60.0	62.2	60.4	52.0	39.7	26.7	21.5	(31)
(32)			MDCB	23.5	26.8	36.0	53.8	67.7	73.3	74.2	69.2	60.1	44.2	28.0	22.4	(32)
(33)		10%	WB	17.9	20.7	27.1	40.3	51.2	58.4	60.8	58.5	50.5	36.7	22.3	16.9	(33)
(34)			MDCB	18.8	22.1	31.4	50.3	63.7	71.1	71.7	65.9	57.6	40.1	23.4	17.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	15.4	20.0	25.8	21.9	21.5	19.4	17.7	16.9	17.5	13.8	14.1	14.9	(35)
(36)			MCDBR	17.9	20.8	28.6	26.7	27.5	25.2	24.0	24.1	24.1	19.2	14.3	19.4	(36)
(37)			MCWBR	16.3	18.1	21.5	14.5	11.9	9.9	9.8	10.8	12.6	12.4	12.8	17.9	(37)
(38)		5% WB	MCDBR	18.1	20.0	26.5	25.6	24.3	21.8	21.1	20.4	20.8	17.8	14.0	19.6	(38)
(39)			MCWBR	16.4	17.5	19.9	14.2	10.6	9.2	9.2	9.7	11.4	11.8	12.5	18.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.171	0.247	0.252	0.308	0.345	0.381	0.381	0.400	0.310	0.263	N/A	0.177	(40)	
(41)		taud	1.943	2.204	2.306	2.291	2.426	2.336	2.362	2.294	2.540	2.459	N/A	1.943	(41)	
(42)		Ebn at Noon	123	213	270	275	274	264	260	240	249	212	N/A	37	(42)	
(43)		Edh at Noon	7	17	24	32	31	35	33	32	20	13	N/A	3	(43)	
(44)	All-Sky Solar Radiation	RadAvg	50	270	841	1453	1668	1758	1521	1160	730	298	75	15	(44)	
(45)		RadStd	4	22	53	92	123	134	106	113	89	33	10	1	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.30	N/A	N/A	N/A	N/A	N/A	-486	-450	N/A	N/A
(47)	Regional (5 neighbors)	+1.55	N/A	N/A	N/A	N/A	N/A	-587	-637	N/A	N/A

Nomenclature: See separate page